

Solid-State Electronics

Chapter 3 & 4 Homework

2024-12-26



Homework 5



3.8 Photoelectric effect and quantum efficiency Cesium metal is to be used as the photocathode material in a photoemissive electron tube because electrons are relatively easily removed from a cesium surface. The work function of a clean cesium surface is 1.9 eV.

a. What is the longest wavelength of radiation which can result in photoemission?

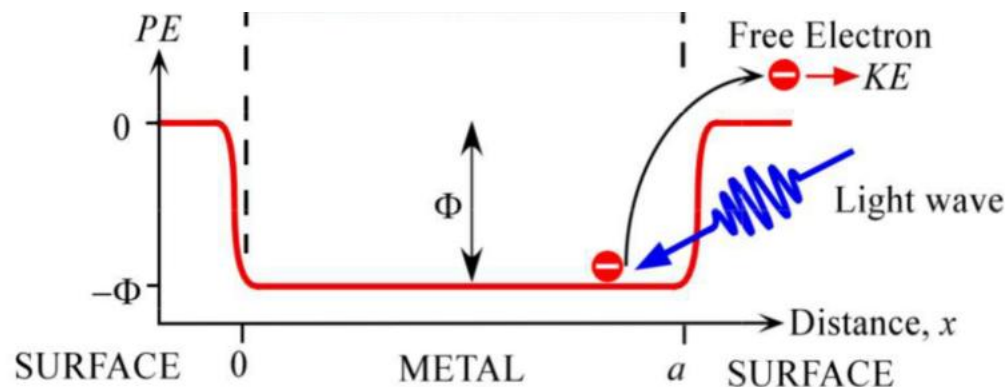
Solution

P168 E.g. 3.2

a. The longest wavelength will be $\Phi = h\nu_o = hc/\lambda_{\max}$

$$\lambda_{\max} = \frac{hc}{\Phi} = \frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{(1.9 \times 1.602 \times 10^{-19} \text{ J})}$$

$$= 6.531 \times 10^{-7} \text{ m} = 653.1 \text{ nm}$$



b. If blue radiation of wavelength **450 nm** is incident onto the Cs photocathode, what will be **the kinetic energy** of the photoemitted electrons in eV? What should be the **voltage** required on the opposite electrode to extinguish the external photocurrent?

Solution

b. The energy of the photon at 450 nm is

$$E_{ph} = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{(450 \times 10^{-9} \text{ m})}$$
$$= 4.417 \times 10^{-19} \text{ J} = 2.76 \text{ eV}$$

The excess energy over Φ goes to kinetic energy of the photoelectron. Thus the electron kinetic energy is

$$KE = E_{ph} - \Phi = (2.76 \text{ eV}) - (1.9 \text{ eV}) = 0.86 \text{ eV}$$

and the stopping voltage for this wavelength will be **-0.86 V**.



c. Quantum efficiency (QE) of a photocathode is defined by,

$$\text{Quantum efficiency} = \frac{\text{Number of photoemitted electrons}}{\text{Number of incident photons}}$$

QE is 100% if each incident photon ejects one electron. Suppose that blue light of wavelength **450 nm** with an intensity of **30 mW cm⁻²** is incident on a Cs photocathode that is a circular disk of diameter **6 mm**. If the emitted electrons are collected by applying a positive bias voltage to the anode, and the photocathode has a QE of 25%, what will be **the photocurrent**?

Solution

c. The number of photons arriving per unit area per unit time at the photocathode is

$$\Gamma_{ph} = \frac{I}{E_{ph}} = \frac{(30 \times 10^{-3} \times 10^4 \text{ J s}^{-1} \text{ m}^{-2})}{(4.417 \times 10^{-19} \text{ J})} = 6.792 \times 10^{20} \text{ s}^{-1} \text{ m}^{-2}$$

The current density is then simply

$$J = e \Gamma_{ph} QE = (1.602 \times 10^{-19} \text{ C})(6.792 \times 10^{20} \text{ s}^{-1} \text{ m}^{-2})(0.25) = 27.2 \text{ A m}^{-2}$$

The photoelectric current is then

$$I = AJ = \frac{\pi d^2}{4} J = \frac{\pi (12 \times 10^{-3} \text{ m})^2}{4} (27.2 \text{ A m}^{-2}) = 3.076 \times 10^{-3} \text{ A} = 3.076 \text{ mA}$$



3.10 Planck's law and photon energy distribution of radiation Planck's law, stated in Equation 3.9, provides the spectral distribution of **the black body radiation intensity in terms of wavelength through I_λ** , intensity per unit wavelength. Suppose that we wish to find the distribution in terms of frequency ν or photon energy $h\nu$. Frequency $\nu = c/\lambda$ and the wavelength range λ to $\lambda + d\lambda$ corresponds to a frequency range ν to $\nu + d\nu$. ($d\lambda$ and $d\nu$ have opposite signs since ν increases as λ decreases.) The intensity $I_\lambda d\lambda$ in λ to $\lambda + d\lambda$ must be the same as the intensity in ν to $\nu + d\nu$, which we can write as $I_\nu d\nu$ where **I_ν is the radiation intensity per unit frequency**. Thus,

$$I_\nu = I_\lambda \left| \frac{d\lambda}{d\nu} \right|$$

The magnitude sign is needed because $\lambda = c/\nu$ results in a negative $d\lambda/d\nu$, and I_ν must be positive by definition. We can simply substitute $\lambda = c/\nu$ for λ in I_λ and obtain I_λ as a function of ν , and then find $|d\lambda/d\nu|$ to find I_ν from the preceding expression.

a. Show that

$$I_\nu = \frac{2\pi (h\nu)^3}{c^2 h^2 [\exp(h\nu / kT) - 1]}$$



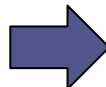
$$I_{\lambda} = \frac{2\pi hc^2}{\lambda^5 \left[\exp\left(\frac{hc}{\lambda kT}\right) - 1 \right]}$$

$$I_{\nu} = I_{\lambda} \left| \frac{d\lambda}{d\nu} \right| = \frac{2\pi hc^2}{\lambda^5 \left[\exp\left(\frac{hc}{\lambda kT}\right) - 1 \right]} \left| \frac{d\lambda}{d\nu} \right|$$

$$\lambda = c / \nu$$

$$d\lambda / d\nu = - c / \nu^2$$

$$\left| d\lambda / d\nu \right| = c / \nu^2$$

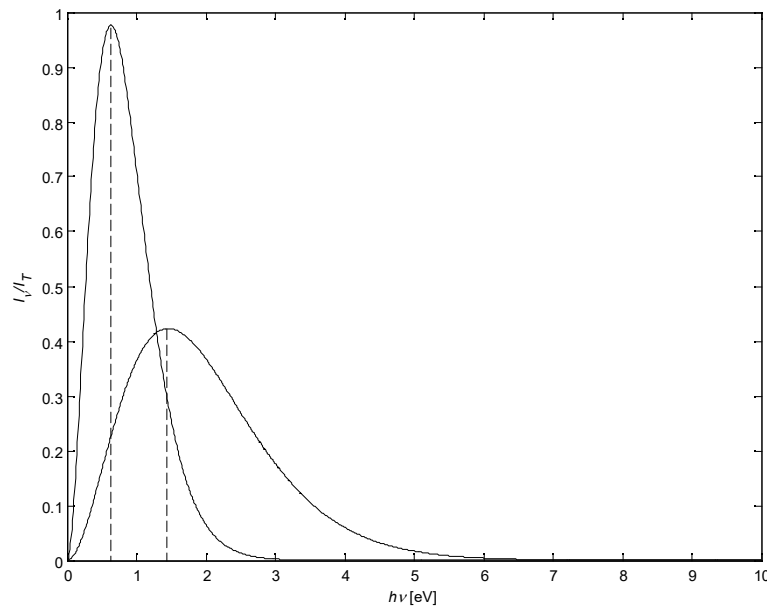


$$I_{\nu} = \frac{2\pi (h\nu)^3}{c^2 h^2 \left[\exp\left(\frac{h\nu}{kT}\right) - 1 \right]}$$



b. If we integrate I_ν over all photon energies (numerically on a calculator or a computer from 0 to say 6 eV), we would obtain the total intensity at a temperature T . Find the total intensity I_T emitted at $T = 2600$ K (a typical incandescent light bulb filament temperature) and at 6000 K (roughly representing the sun's spectrum). Plot $y = I_\nu/I_T$ versus the photon energy in eV. What are **the photon energies** for the peaks in the distributions? Calculate the corresponding wavelength for each using $\lambda = c/\nu$ and then compare these wavelengths with those predicted by Wien's law, $\lambda_{\max}T \approx 2.89 \times 10^{-3}$ m K.

Solution



By integrating numerically using any math software, the total intensity

$$\begin{aligned} I_\nu(2600 \text{ K}) &= 1.07 \times 10^{-7} / \text{h} \\ &= 1.615 \times 10^{25} \text{ eV/m}^2 \cdot \text{s} \\ &= 2.588 \times 10^6 \text{ W/m}^2 \end{aligned}$$

and

$$\begin{aligned} I_\nu(6000 \text{ K}) &= 3.03 \times 10^{-7} / \text{h} \\ &= 4.567 \times 10^{26} \text{ eV/m}^2 \cdot \text{s} \\ &= 7.318 \times 10^7 \text{ W/m}^2 \end{aligned}$$



At 2600 K: at the peak in intensity, we get

$$h\nu = 0.63 \text{ eV or } hc/\lambda = 0.63 \text{ eV}$$

or $\lambda_{2600} = (4.135 \times 10^{-15} \text{ eV s})(3 \times 10^8 \text{ ms}^{-1})/(0.63 \text{ eV}) = \mathbf{1.97 \times 10^{-6} \text{ m}}$ or **1.97 μm**

At 6000 K: at the peak in intensity, we get

$$h\nu = 1.46 \text{ eV or } hc/\lambda = 1.46 \text{ eV}$$

or $\lambda_{6000} = (4.135 \times 10^{-15} \text{ eV s})(3 \times 10^8 \text{ ms}^{-1})/(1.46 \text{ eV}) = \mathbf{8.5 \times 10^{-7} \text{ m}}$ or **850 nm**

Using Wien's law,

$$\lambda_{\text{max}} T = 2.89 \times 10^{-3} \text{ m K}$$

$$\lambda_{\text{max}} = (2.89 \times 10^{-3} \text{ m K}/2600 \text{ K}) = 1.11 \times 10^{-6} \text{ m or } \mathbf{1.11 \mu\text{m}}$$
 at 2600 K

and

$$\lambda_{\text{max}} = (2.89 \times 10^{-3} \text{ m K}/6000 \text{ K}) = 4.8 \times 10^{-7} \text{ m or } \mathbf{480 \text{ nm}}$$
 at 6000 K

Temperature (T)	λ_{max} (Planck's law)	λ_{max} (Wien's law)
2600	1.97 μm	1.11 μm
6000	850 nm	480 nm

Homework 6



3.13 Heisenberg's uncertainty principle Show that if the uncertainty in the position of a particle is on the order of its de Broglie wavelength, then the uncertainty in its momentum is about the same as the momentum value itself.

Solution

The de Broglie wavelength is $\lambda = \frac{h}{p}$

where p is the momentum. From Heisenberg's uncertainty principle we have,

$$\Delta x \Delta p \approx \hbar \quad \text{P182 Eq. 3.24}$$

If we take the uncertainty in the position to be of the order of the wavelength, $\Delta x \sim \lambda$, then

$$\Delta p \approx \frac{\hbar}{\Delta x} \approx \left(\frac{1}{2\pi} \right) \frac{h}{\lambda} = \left(\frac{1}{2\pi} \right) p \sim p$$

so that the uncertainty in the momentum will be of the same order as the momentum itself.



3.15 Tunneling

a. Consider the phenomenon of tunneling through **a potential energy barrier of height V_0 and width a** , as shown in Figure 3.16. What is **the probability** that the electron will be reflected? Given the transmission coefficient T , can you find **the reflection coefficient R** ? What happens to R as a or V_0 or both become very large?

The relative probability of the electron from region I through to III, transmission coefficient T

P186 Eq. 3.29

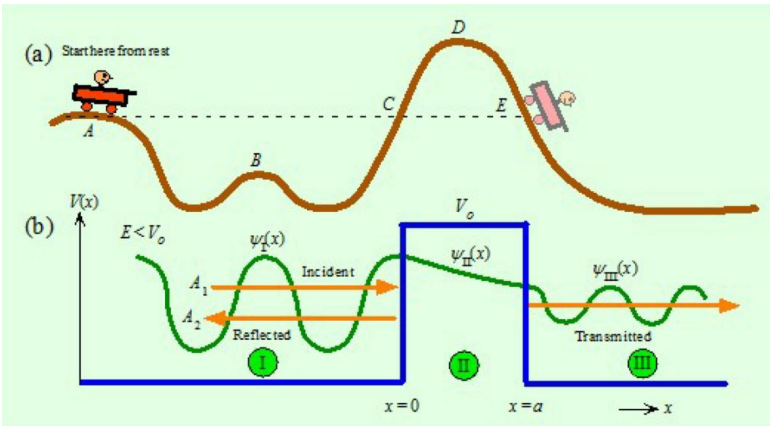
$$T = \frac{|\varphi_{III}(x)|^2}{|\varphi_I(incident)|^2} = \frac{C_1^2}{A_1^2} = \frac{1}{1 + D \sinh^2(a\alpha)}$$

where $\begin{cases} \sinh(x) = \frac{e^x - e^{-x}}{2} \\ D = \frac{V_0^2}{4E(V_0 - E)} \end{cases}$

For a wide or high barrier, using $a\alpha \geq 1 : \sinh(a\alpha) \approx \exp(a\alpha)/2$

$$T = T_0 \exp(-2a\alpha) \quad \text{where} \quad T_0 = \frac{16E(V_0 - E)}{V_0^2}$$

$$T = 16 \left(\frac{E}{V_0} \right) \left(1 - \frac{E}{V_0} \right) e^{-\sqrt{\frac{8m}{\hbar^2}(V_0 - E)}a}$$



Solution

a. The relative reflection probability or reflection coefficient R is given as the ratio of the square of the amplitude of the reflected wave to that of the incident wave, which is:

P186 Eq. 3.33; Eq. 3.29

$$R = \frac{A_2^2}{A_1^2} = 1 - T \qquad T = \frac{1}{1 + D(\sinh[a\alpha])^2}$$

where a is the width of the potential energy barrier, α is the rate of decay, and D is given by:

$$D = \frac{V_0^2}{4E(V_0 - E)}$$

To determine the behavior of R as a or V_0 or both become very large, we can use the equation $R = 1 - T$ to express R in terms of a and D (remember D is a function of V_0).

$$R = 1 - T = 1 - \frac{1}{1 + D(\sinh[a\alpha])^2} = \frac{D(\sinh[a\alpha])^2}{1 + D(\sinh[a\alpha])^2}$$

As a or V_0 becomes large, $R = 1$ for total reflection.

Solution

b. For a wide barrier ($\alpha a \gg 1$), show that T_0 can at most be 4 and that $T_o = 4$ when $E = \frac{1}{2} V_0$.

We need to find the maximum value of T_0 . Since T_0 depends on the energy E , we can differentiate it with respect to E , set the result to 0 and isolate E .

$$T_0 = \frac{16E(V_0 - E)}{V_0^2}$$
$$\frac{dT_o}{dE} = 16 \left(\frac{-2E + V_o}{V_o^2} \right) = 0$$

P186 Eq. 3.32

$$E = \frac{1}{2} V_o$$

Thus T_o is maximum when $E = V_0/2$. If this expression for energy is substituted back into the equation for T_0 to find its maximum value (T_o'):

$$T_o' = \frac{16E(V_o - E)}{V_o^2} = \frac{16 \left(\frac{1}{2} V_o \right) \left(V_o - \frac{1}{2} V_o \right)}{V_o^2} = \frac{16V_o^2}{4V_o^2} = 4$$

Homework 7



3.27 Hund's rule The carbon atom has the electronic structure $2s^2 2p^2$ in its ground state. The ground state and various possible excited states of C are shown in Figure 3.47. The following energies are known for the states *a* to *e* in Figure 3.47, not in any particular order: 0, 7.3 eV, 4.1 eV, 7.9 eV, and 1.2 eV. Using reasonable arguments match these energies to the states *a* to *e*. Use Hund's rule to establish the ground state with 0 eV. If you have to flip a spin to go from the ground to another configuration, that would cost energy. If you have to move an electron from a lower *s* to *p* or from *p* to a higher *s*, that would cost a lot of energy. Two electrons in the same orbital (obviously with paired electrons) would have substantial Coulombic repulsion energy.

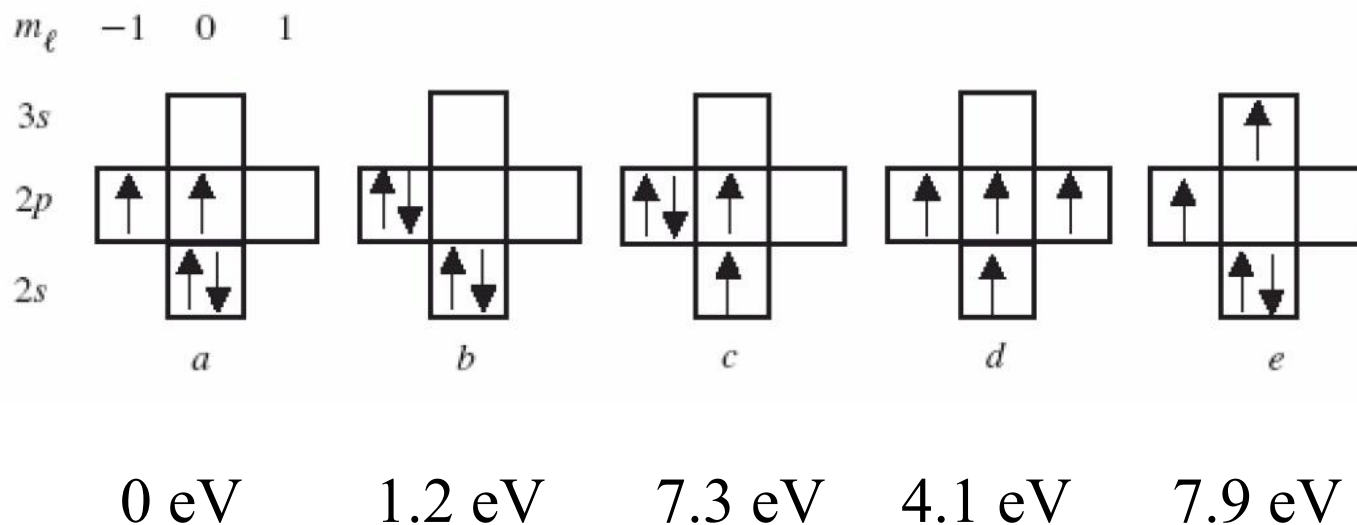
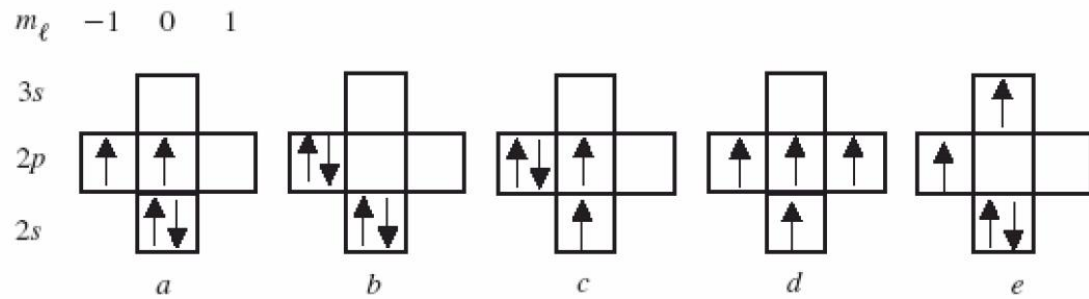


Figure 3.47

Some possible states of the carbon atom, not in any particular order.





Solution

1. According to the Hund's rule, a is the standard state, i.e. $E_a = 0$ eV.
2. Compared with a , two electrons are in $2p$ orbital in b , resulting in Coulomb energy, but it is lower than the energy required to move the electrons to a higher level or spin flip, i.e. $E_b > E_a$.
3. Compared with b , one electron in the $2s$ orbital of c is transitioned to the $2p$ orbital, and with paired electrons, results in greater energy, $E_c > E_b$.
4. Compared with c , d adheres more closely to Hund's rule, resulting in lower energy, i.e. $E_d < E_c$.
5. When comparing b to d , d has more electrons in $2p$, leading to $E_b < E_d$.
6. Compared to c , move one electron to $3s$ in e , i.e. $E_e > E_c$.

Homework 8



4.11 X-ray emission spectrum from sodium

Structure of Na atom is $[\text{Ne}]3s^1$. Figure 4.59a shows formation of the $3s$ and $3p$ energy bands in Na as a function of internuclear separation. Figure 4.59b shows the X-ray emission spectrum (called the L-band) from crystalline sodium in the soft x-ray range as explained in Example 4.6.

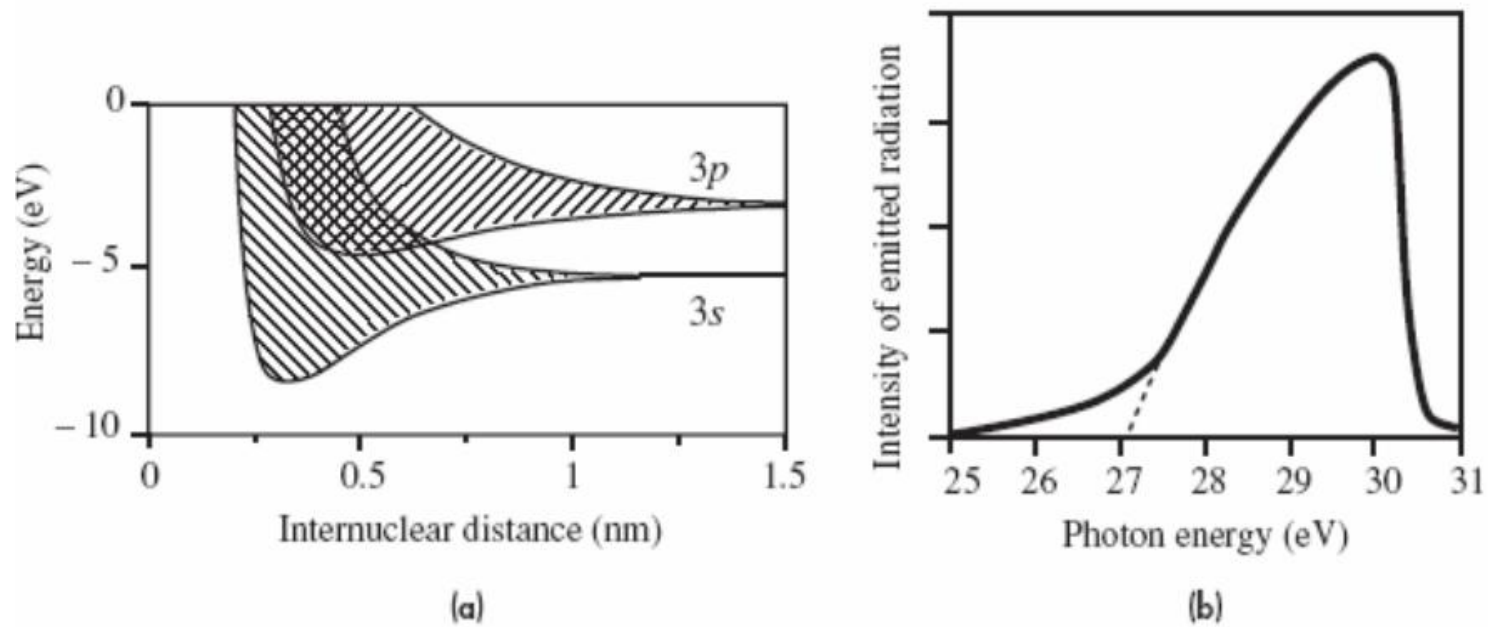


Figure 4.59
(a) Energy band formation in sodium.
(b) L-emission band of X-rays from sodium.
| SOURCE: [b] Data extracted from W. M. Cadt and D. H. Tomboulion, *Phys. Rev.*, **59**, 1941, p. 381.

a. From Figure 4.59a, estimate the nearest neighbor equilibrium separation between Na atoms in the crystal if some electrons in the 3s band spill over into the states in the 3p band.

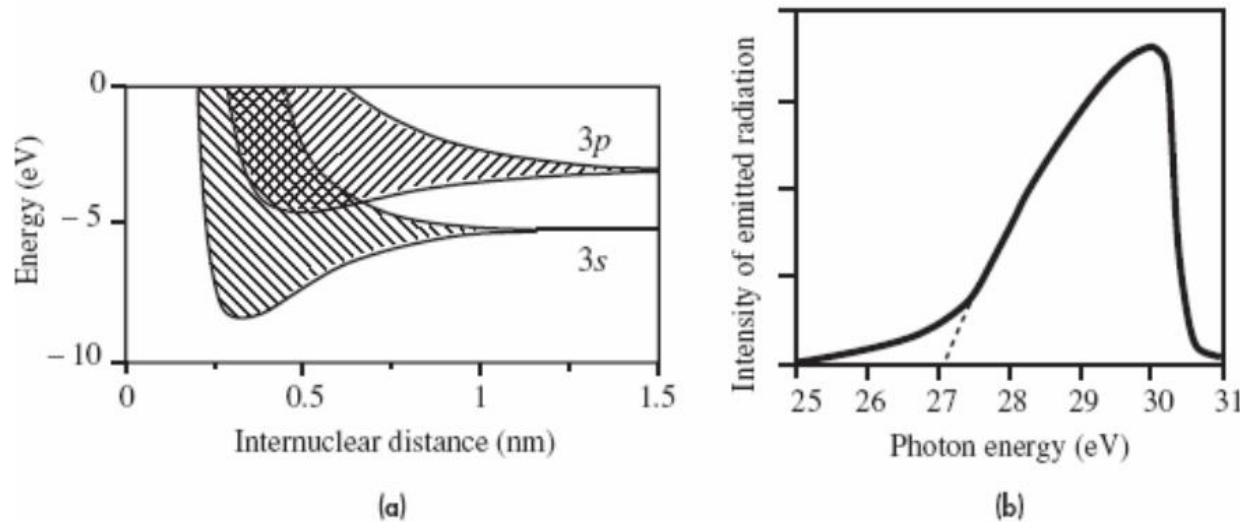


Figure 4.59

(a) Energy band formation in sodium.

(b) L-emission band of X-rays from sodium.

| SOURCE: [b] Data extracted from W. M. Cadt and D. H. Tomboulion, *Phys. Rev.*, **59**, 1941, p. 381.

Solution

a. As represented in Figure 4.59a, the estimated nearest interatomic separation is near the minimum of the 3s band, or slightly above it, 0.36 - 0.37 nm.

b. **Explain** the origin of X-ray emission band in Figure 4.59b and the reason for calling it the *L*-band.

P256 E.g. 4.6

Solution

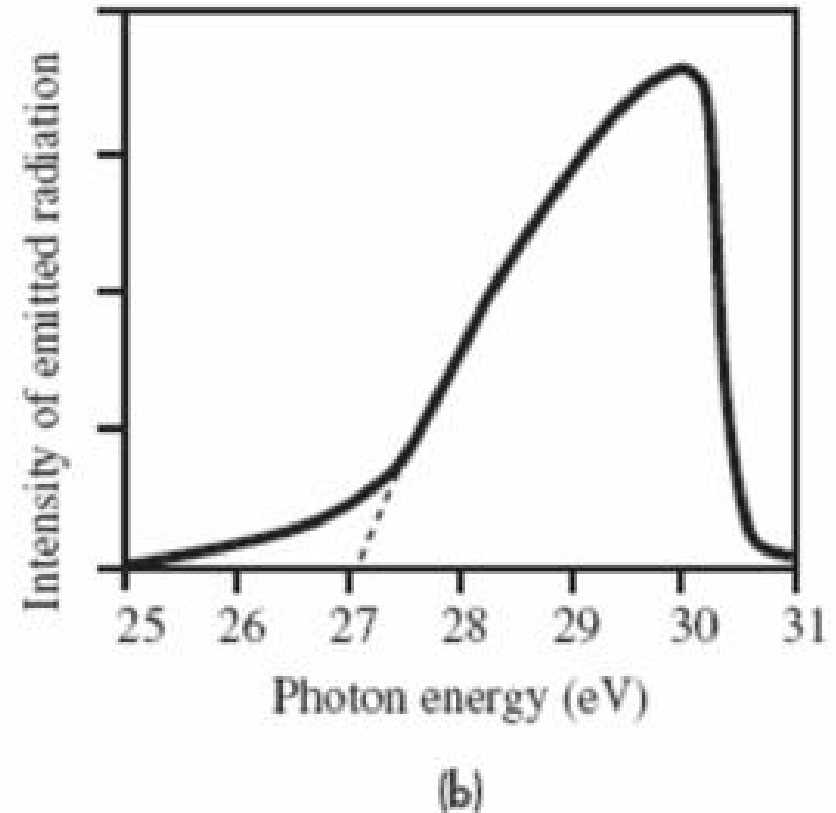
When an electron, for some reason, is **leaving** the closed inner *L*-shell of an atom, an empty state is **created** there. An electron from the energy band of the metal **drops** into the *L*-shell to fill the vacancy and **emits** a soft X-ray photon in this process. The spectrum of this X-ray emission from metal involves a range of energies, corresponding to transitions from the bottom of the band and from the Fermi level to the *L*-shell. So all the X-ray photons emitted from the electrons during their transitions to the *L*-shell will have energies lying in that range (band) and because all of them are emitted due to a transition to the *L*-shell, this X-ray band is called *L*-band.



c. What is the Fermi energy of the electrons in Na from Figure 4.59b?

Solution

As explained in part b, the width of the X-ray band corresponds to **the distance from the bottom of the energy band to the Fermi level**. As shown in Figure 4.59b, the position of the Fermi level with respect to the bottom of the energy band is approximately 3.2 eV.



d. Taking the valency of Na to be I, what is the expected Fermi energy and how does it compare with that in (c)?

P261 Eq. 4.22

Solution

Since the temperature dependence of the Fermi level is really very weak, we can neglect it. Then $E_F(T) = E_{F0}$ and using Equation 4.22, we receive

$$E_F = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}}$$

The electron concentration in Na can be calculated, assuming that each Na atom donates exactly one electron to the crystal. Taking from Appendix B the density of Na d (0.97 g cm^{-3}) and its atomic mass M_{at} (22.99 g mol^{-1}), we receive

$$n = \frac{d N_A}{M_{at}} = \frac{(0.97 \text{ g cm}^{-3}) (6.022 \times 10^{23} \text{ mol}^{-1})}{(22.99 \text{ g mol}^{-1})} = 2.54 \times 10^{22} \text{ cm}^{-3} = 2.54 \times 10^{28} \text{ m}^{-3}$$

Thus the Fermi level is

$$E_F = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}} = \frac{(6.626 \times 10^{-34} \text{ J s})^2}{8(9.1 \times 10^{-31} \text{ kg})} \left[\frac{3(2.54 \times 10^{28} \text{ m}^{-3})}{\pi} \right]^{\frac{2}{3}} = 5.05 \times 10^{-19} \text{ J}$$

$E_F = 3.15 \text{ eV}$, which is very close to the value obtained from the X-ray spectrum.



Homework 9



4.9 Fermi energy and electron concentration Consider the metals in Table 4.8 from groups I, II and III in the Periodic Table. Calculate the Fermi energies at absolute zero, and compare the values with the experimental values. What is your conclusion?

Table 4.8					
Metal	Group	M_{at}	Density (g cm ⁻³)	E_F (eV) [Calculated]	E_F (eV) [Experiment]
Cu	I	63.55	8.96	—	6.5
Zn	II	65.38	7.14	—	11.0
Al	III	27	2.70	—	11.8

Solution

Since Cu is in group I, its valency is also 1. The electron concentration n is then the atomic concentration multiplied by the group number, or:

$$n = (\text{Valency}) \frac{N_A D}{M_{at}} = (1) \frac{(6.022 \times 10^{23} \text{ mol}^{-1})(8.96 \times 10^3 \text{ kg/m}^3)}{63.55 \times 10^{-3} \text{ kg/mol}} = 8.490 \times 10^{28} \text{ m}^{-3}$$

$$E_{FO} = \frac{h^2}{8m_e} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}} \left(\frac{1}{q} \right)$$

$$E_{F0} = \frac{(6.626 \times 10^{-34} \text{ J s})^2}{8(9.109 \times 10^{-31} \text{ kg})} \left(\frac{3(8.490 \times 10^{28} \text{ m}^{-3})}{\pi} \right)^{\frac{2}{3}} \left(\frac{1}{1.602 \times 10^{-19} \text{ J/eV}} \right)$$

$$E_{FO} = 7.04 \text{ eV}$$

Comparing with the experimental value:

$$|\% \text{ difference}| = \left| \frac{7.04 \text{ eV} - 6.5 \text{ eV}}{6.5 \text{ eV}} \right| \times 100\% = 8.31\%$$

E_{FO} can be calculated for Zn and Al in the same way (remember to take into account the different valencies). The values are summarized in the following table and it can be seen that calculated values are close to experimental values:

Metal	$n \text{ (m}^{-3}\text{)} (\times 10^{28})$	$E_{FO} \text{ (eV)}$ (calculated)	$E_{FO} \text{ (eV)}$ (experimental)	%difference
Cu	8.490	7.04	6.5	8.31
Zn	13.15	9.43	11.0	14.3
Al	18.07	11.7	11.8	0.847



4.10 Temperature dependence of the Fermi energy

a. Given that the Fermi energy for Cu is 7.0 eV at absolute zero, calculate **the E_F at 300 K**. What is the percentage change in E_F and what is your conclusion?

Solution

a. The Fermi energy in eV at 0 K is given as 7.0 eV.

$$E_F = E_{F0} \left(1 - \frac{\pi^2}{12} \left[\frac{kT}{E_{F0}} \right]^2 \right) \quad \text{P262 Eq. 4.23}$$

$$E_F = (7.0 \text{ eV}) \left(1 - \frac{\pi^2}{12} \left[\frac{(1.381 \times 10^{-23} \text{ J/K})(300 \text{ K})}{(7.0 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} \right]^2 \right) = \mathbf{6.999921 \text{ eV}}$$

$$|\% \text{ difference}| = \left| \frac{6.999921 \text{ eV} - 7.0 \text{ eV}}{7.0 \text{ eV}} \right| \times 100\% = 0.00129\%$$



b. Given the Fermi energy for Cu at absolute zero, calculate **the average energy and mean speed per conduction electron** at absolute zero and 300 K, and comment.

Solution

b. The average energy per electron at 0 K is:

$$E_{av}(0 \text{ K}) = 3/5 (E_{F0}) = \mathbf{4.2 \text{ eV}} \quad \text{P262 Eq. 4.25}$$

The average energy at 300 K can be calculated:

$$E_{av}(T) = \frac{3}{5} E_{F0} \left(1 + \frac{5\pi^2}{12} \left[\frac{kT}{E_{FO}} \right]^2 \right) \quad \text{P262 Eq. 4.26}$$

$$E_{av}(300 \text{ K}) = \frac{3}{5} (7.0 \text{ eV}) \left(1 + \frac{5\pi^2}{12} \left[\frac{(1.381 \times 10^{-23} \text{ J/K})(300 \text{ K})}{(7.0 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} \right]^2 \right)$$

$$E_{av}(300 \text{ K}) = \mathbf{4.200236 \text{ eV}}$$

This is a very small change.



Assume that the mean speed will be close to the effective speed v_e . Effective speed at absolute zero is denoted as v_{eo} , and is given by:

$$E_{av}(0\text{ K}) \times q = \frac{1}{2} m_e v_{eo}^2$$

$$v_{eo} = \sqrt{2 \frac{q E_{av}(0\text{ K})}{m_e}} = \sqrt{2 \frac{(1.602 \times 10^{-19} \text{ J/eV})(4.2 \text{ eV})}{(9.109 \times 10^{-31} \text{ kg})}} = \mathbf{1215446 \text{ m/s}}$$

At 300 K, the effective speed is v_e :

$$v_e = \sqrt{2 \frac{q E_{av}(300\text{ K})}{m_e}} = \sqrt{2 \frac{(1.602 \times 10^{-19} \text{ J/eV})(4.200236 \text{ eV})}{(9.109 \times 10^{-31} \text{ kg})}} = \mathbf{1215480 \text{ m/s}}$$

Comparing the values:

$$|\% \text{ difference}| = \left| \frac{1215480 \text{ m/s} - 1215446 \text{ m/s}}{1215446 \text{ m/s}} \right| \times 100\% = \mathbf{0.002797\%}$$



THANKS !

